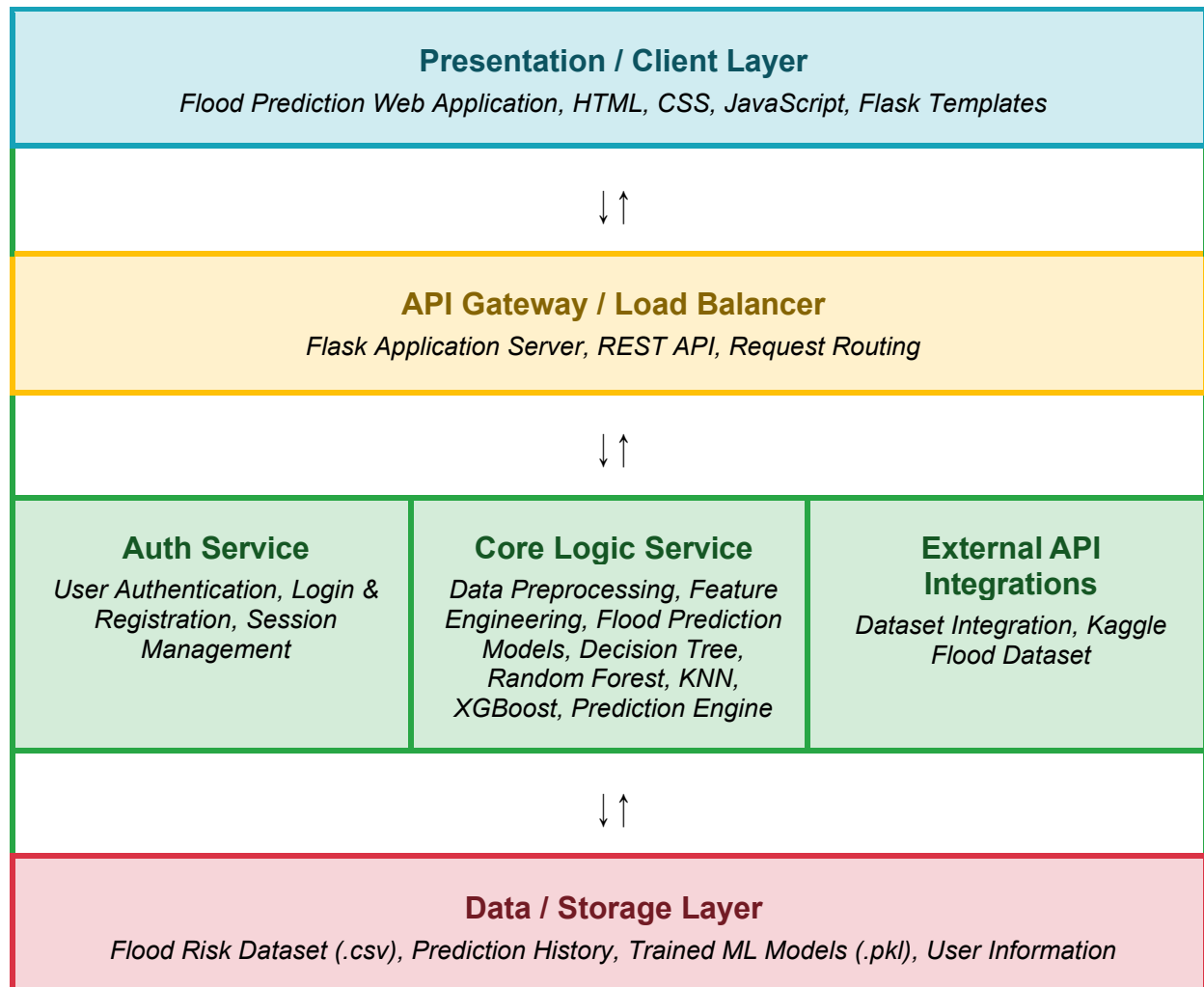


Solution Architecture

Date	26 June 2026
Team ID	
Project Name	Rising Waters – A Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering weather parameters and displaying flood prediction results.	HTML, CSS, JavaScript, Flask Templates
API Gateway	Receives client requests, routes them to backend services, and returns prediction results.	Flask Application Server, REST API
Microservices	Performs data preprocessing, feature engineering, model training, and flood risk prediction using ML algorithms.	Python, Scikit-learn, Pandas, NumPy, XGBoost
Database	Stores flood dataset, trained machine learning models, user information, and prediction history.	CSV Dataset, Pickle (.pkl) Models